Arista Access Node ELEC-ACC-01 QoS Congestion Event Troubleshooting and Resolution SOP

Node: ELEC-ACC-01 | **Vendor:** Arista | **Severity:** Critical | **Alarm Code:** ALARM_1392

Description

The Access Node ELEC-ACC-01, situated at the Electronic City Hub, exhibits a Critical QoS congestion event, manifesting as Microbursts, interface tail drops, and QoS queue starvation. This phenomenon originates from the intricate dynamics of Quality of Service (QoS) and Congestion Avoidance mechanisms within the Arista EOS (Extensible Operating System). The Arista EOS implements a sophisticated scheduling algorithm, which allocates bandwidth among competing traffic flows based on their assigned QoS policies. However, in scenarios where the aggregate traffic load exceeds the available bandwidth, the system resorts to various congestion avoidance techniques, including Random Early Detection (RED) and Tail Drop. When these mechanisms are overwhelmed, it can lead to Microbursts and interface tail drops, ultimately causing QoS queue starvation. This SOP aims to troubleshoot and resolve the underlying causes of this congestion event, ensuring optimal network performance and QoS for critical applications.

Pre-requisite Checks

- Verify that the redundant link status is operational at the Electronic City Hub.
- Dispatch remote hands to physically access the Node ELEC-ACC-01, if necessary.
- Ensure that the network management system (NMS) is accessible and functioning correctly.

Potential Root Causes

Cause	Probability
Inadequate QoS policy configuration	High
Insufficient network bandwidth or oversubscription	Medium
Malfunctioning or misconfigured Arista EOS features (e.g., RED, Tail Drop)	Low

Diagnostic Commands & Expected Outputs

```
Command: show qos interface ELEC-ACC-01, Expected Output: Interface QoS policy configuration and statistics, Description: This command displays the QoS policy configuration and statistics for the Access Node ELEC-ACC-01, allowing us to assess the QoS policy configuration and identify any potential issues.

Command: show interface ELEC-ACC-01 stats, Expected Output: Interface statistics, including packet drops and errors, Description: This command provides detailed interface statistics, including packet drops and errors, which can help identify signs of congestion or interface issues.
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```
Command: show qos policer ELEC-ACC-01, Expected Output: Policer configuration and statistics, Description: This command displays the policer configuration and statistics for the Access Node ELEC-ACC-01, enabling us to examine the policer settings and identify any potential issues.
```

Resolution Steps

Step 1: Step: Adjust QoS policy configuration, Command: config qos policy ELEC-ACC-01 priority-map, Description: Modify the QoS policy configuration to prioritize critical applications or adjust the priority map to ensure adequate bandwidth allocation.

Step 2: Step: Increase network bandwidth or modify oversubscription ratios, Command: config interface ELEC-ACC-01 bandwidth 1000, Description: Increase the network bandwidth or modify the oversubscription ratios to ensure sufficient capacity for the aggregate traffic load.

Step 3: Step: Reset or reconfigure Arista EOS features (e.g., RED, Tail Drop), Command: config feature red reset, Description: Reset or reconfigure the Arista EOS features to their default settings or modify the configuration to address any potential issues.

Post-Check Verification

```
Command: show qos interface ELEC-ACC-01, Expected Output: Interface QoS policy configuration and statistics, Description: Verify that the QoS policy configuration and statistics are corrected and indicate no signs of congestion or interface issues.

Command: show interface ELEC-ACC-01 stats, Expected Output: Interface statistics, including packet drops and errors, Description: Verify that the interface statistics indicate no signs of packet drops or errors, confirming that the congestion event has been resolved.

Command: show qos policer ELEC-ACC-01, Expected Output: Policer configuration and statistics, Description: Verify that the policer configuration and statistics indicate no signs of issues, confirming that the policer settings are correct.
```

Roll-back Procedures

- Step: Revert QoS policy configuration changes, Command: rollback qos policy ELEC-ACC-01, Description: Revert the QoS policy configuration changes to their previous state, if necessary.

- Step: Revert bandwidth or oversubscription ratio changes, Command: rollback interface ELEC-ACC-01 bandwidth, Description: Revert the bandwidth or oversubscription ratio changes to their previous state, if necessary.

- Step: Revert Arista EOS feature changes, Command: rollback feature red reset, Description: Revert the Arista EOS feature changes to their default settings or previous state, if necessary.